



Predicting Vasoplegia at Bypass Separation with Three Variables



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KEY FINDINGS

Three numbers already on every perfusion record predict vasoplegia at the moment of separation as well as a 38-parameter machine learning model.

0.858 vs 0.855

AUROC of 3 routine variables vs the 38-parameter elastic net (difference +0.003, 95% CI -0.020 to 0.027)

+0.08 to +0.09

AUROC added by the minute-level flowsheet to the same algorithms on static data (Holm-adjusted p = 0.010)

0.84 vs 0.57

Among 438 patients not yet hypotensive at separation: three-variable model vs terminal MAP alone

256 vs 3,209

Patients needed (Riley) for a pre-specified 3- vs a 38-parameter model; this cohort has 704

1 Background

Normal tone → **Vasoplegia**

Vasoplegia is distributive shock after cardiopulmonary bypass (CPB). Vascular tone collapses and arterial pressure falls while the heart itself pumps normally.

Treatment escalates through vasopressors to rescue agents such as methylene blue, hydroxocobalamin, and angiotensin II.

pressure holds → pressure falls

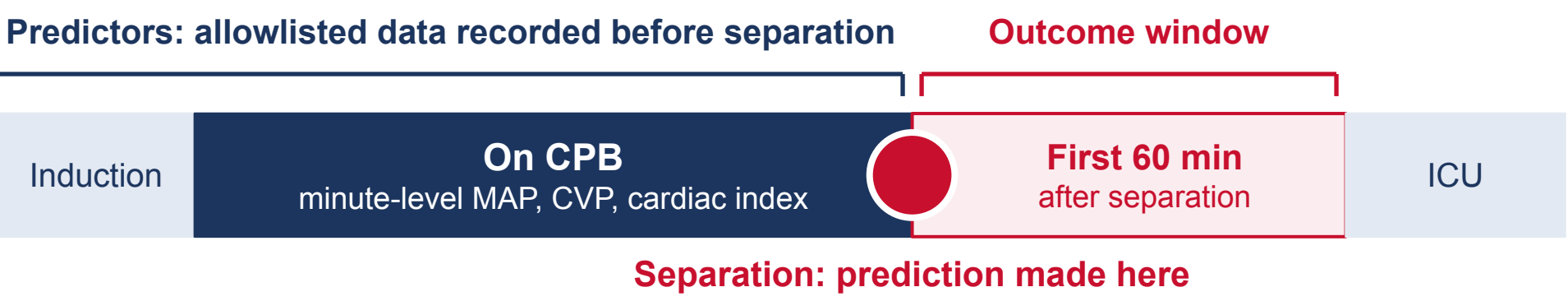
- Reported incidence spans 2.5% to 66%, largely because definitions differ [1, 2].
- Associated with longer intensive care stays and higher mortality [3].
- Usually recognized only after vasopressor needs escalate, although earlier rescue therapy is associated with better survival [4].

Where vasoplegia risk tools operate

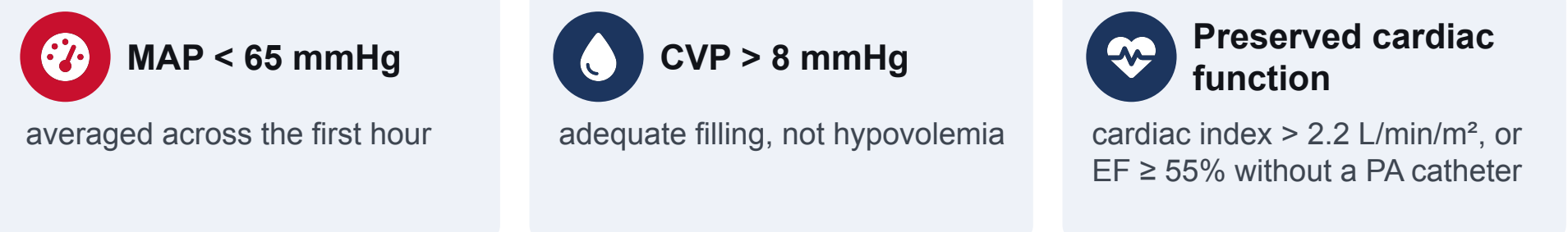


Aim Early on-bypass hypotension was linked to vasoplegia with a single variable in 2009 [7], but no model has measured what the **minute-level intraoperative record** adds at the moment of separation, or how much model complexity a cohort with 45 events can support.

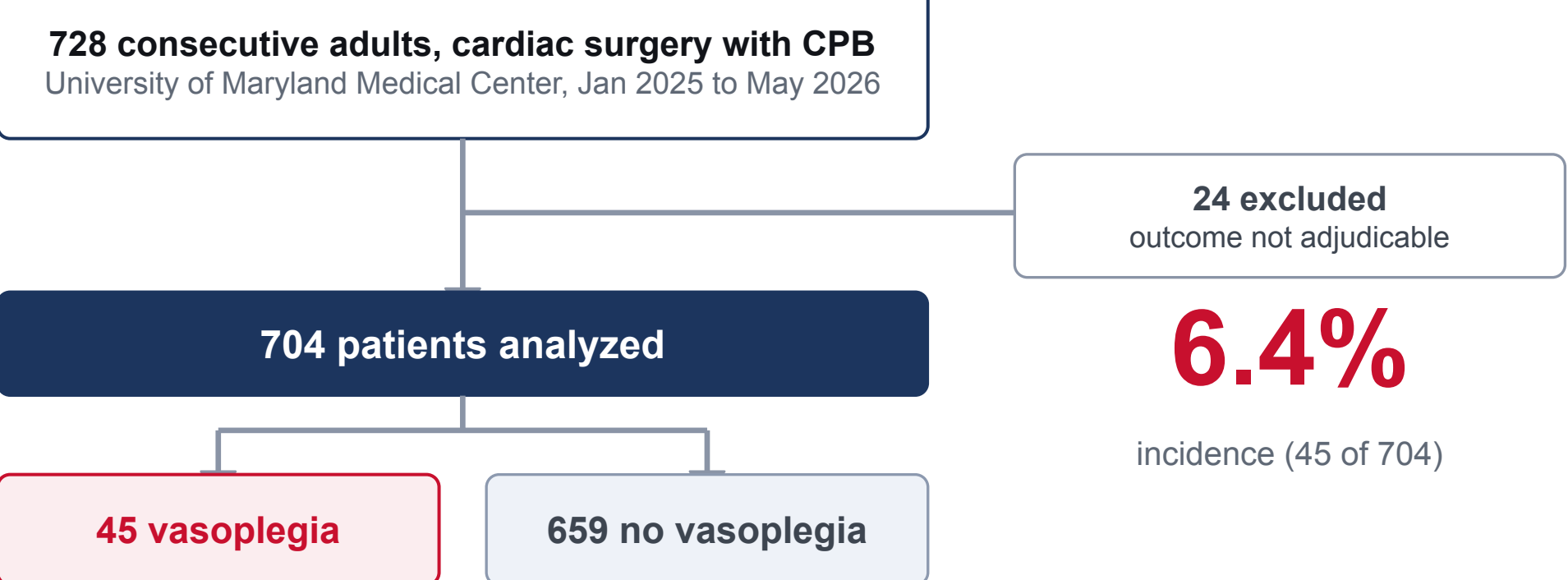
2 Study Population



Outcome, fixed before modeling (all three in the first hour after separation)



Pressure based by design: no vasopressor-dose criterion, so the label is not conditioned on the treatment a deployed score would trigger. **Label check:** systemic vascular resistance index (SVRI) was lower in events (median 1,314 vs 1,784 dyn·s·cm⁻⁵·m², p < 0.001; 338 patients with catheter data, 23 events). SVRI shares the defining pressures, so this confirms consistency only.

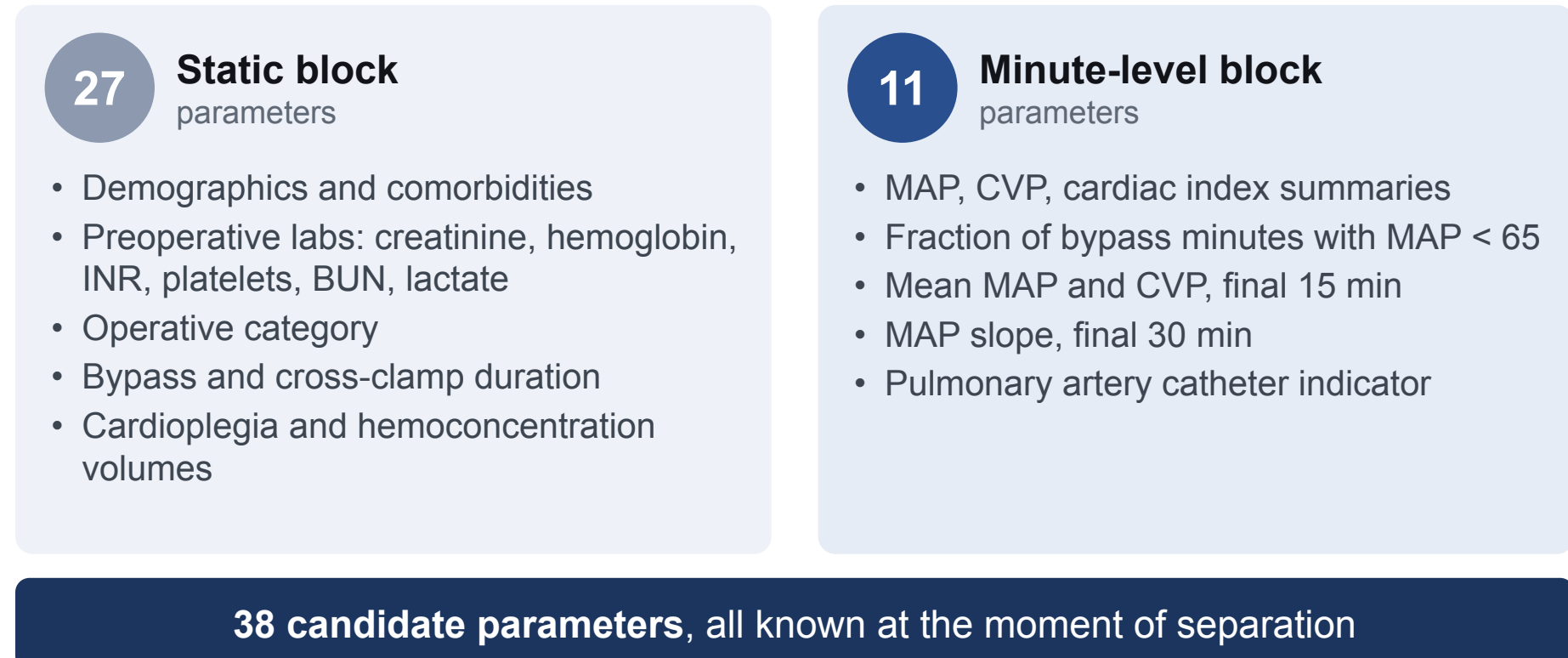


Cohort characteristics by outcome

Characteristic	Vasoplegia (45)	None (659)	p
Age, years	64 [55, 70]	63 [55, 71]	0.75
Male sex	32 (71%)	451 (68%)	0.87
Diabetes	12 (27%)	175 (27%)	>0.99
CABG	16 (36%)	356 (54%)	0.020
Preop hemoglobin, g/dL	10.2 [8.9, 14.0]	13.1 [11.1, 14.4]	0.004
Preop creatinine, mg/dL	1.15 [0.91, 1.43]	0.99 [0.80, 1.20]	0.006
Preop INR	1.10 [1.00, 1.20]	1.00 [1.00, 1.10]	0.007
Aortic surgery	14 (31%)	111 (17%)	0.025
CPB duration, min	187 [146, 222]	116 [90, 156]	<0.001
Cross-clamp duration, min	122 [91, 159]	82 [59, 111]	<0.001
Mean MAP on bypass, mmHg	68.8 [65.2, 72.1]	74.8 [71.2, 78.3]	<0.001
Bypass minutes with MAP < 65	35% [24, 51]	22% [14, 31]	<0.001

Median [IQR] with Mann-Whitney p, or n (%) with Fisher exact p.

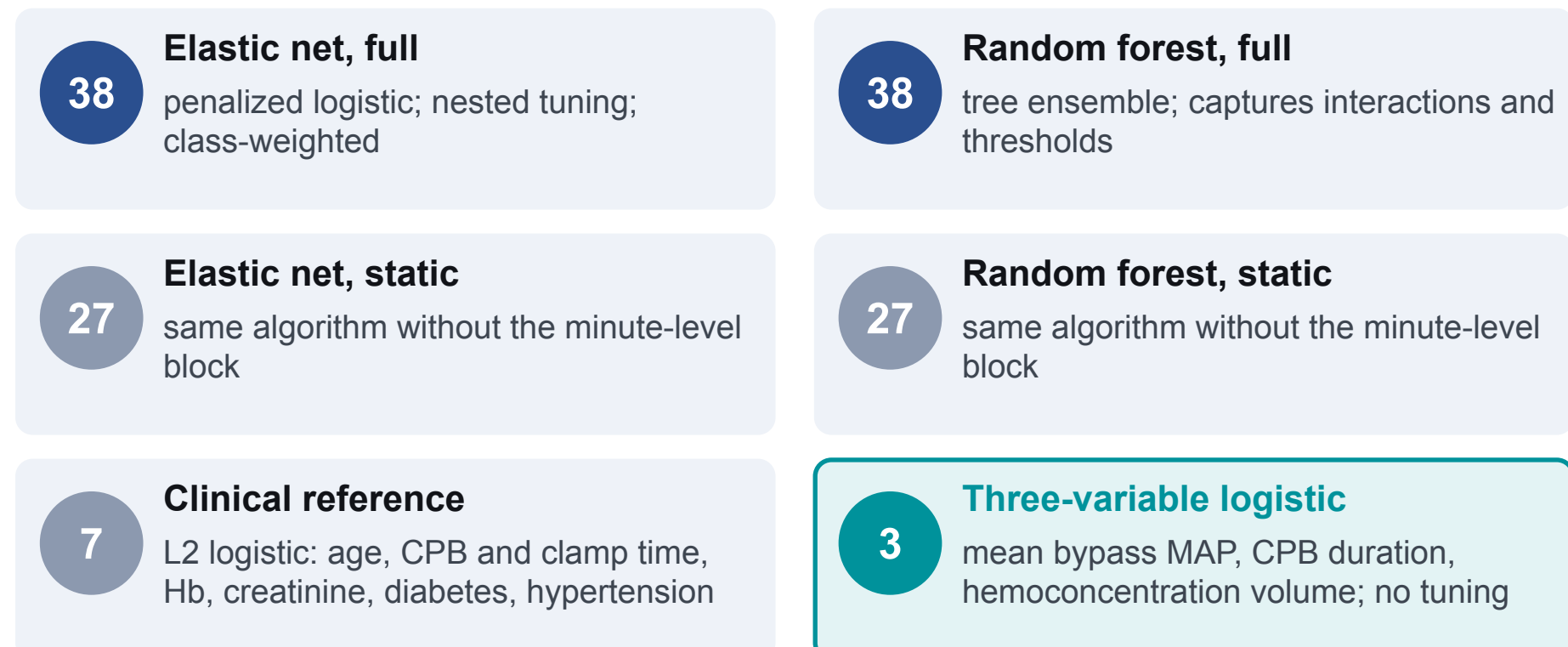
3 Data and Features



38 candidate parameters, all known at the moment of separation

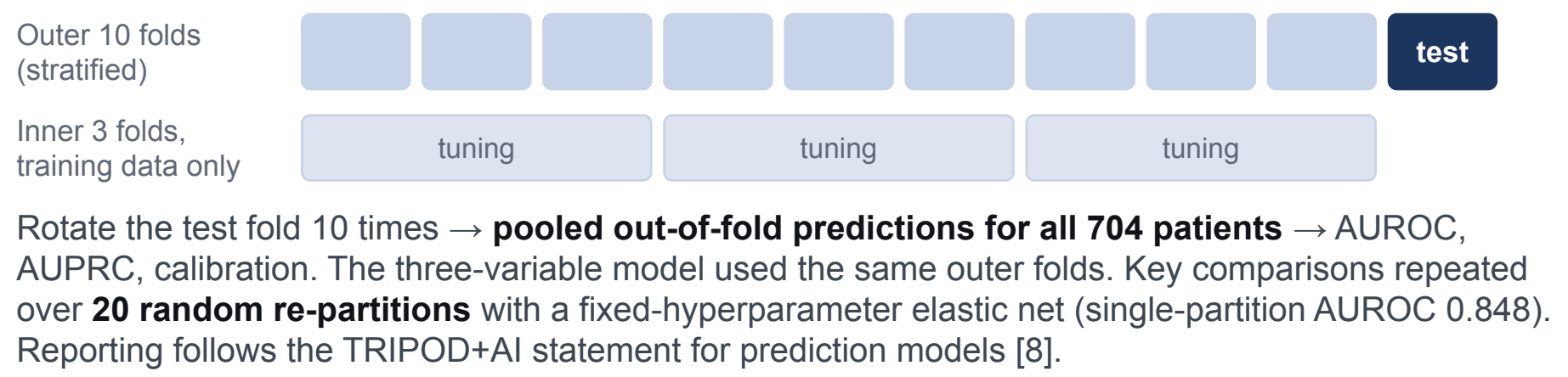
Excluded by allowlist: anything recorded after separation, the outcome-defining first-hour averages, and intraoperative vasopressor or rescue-drug exposures. **Missing values:** training-fold median imputation inside every split (cardiac index 48% missing, preoperative lactate 69%).

4 Model Development



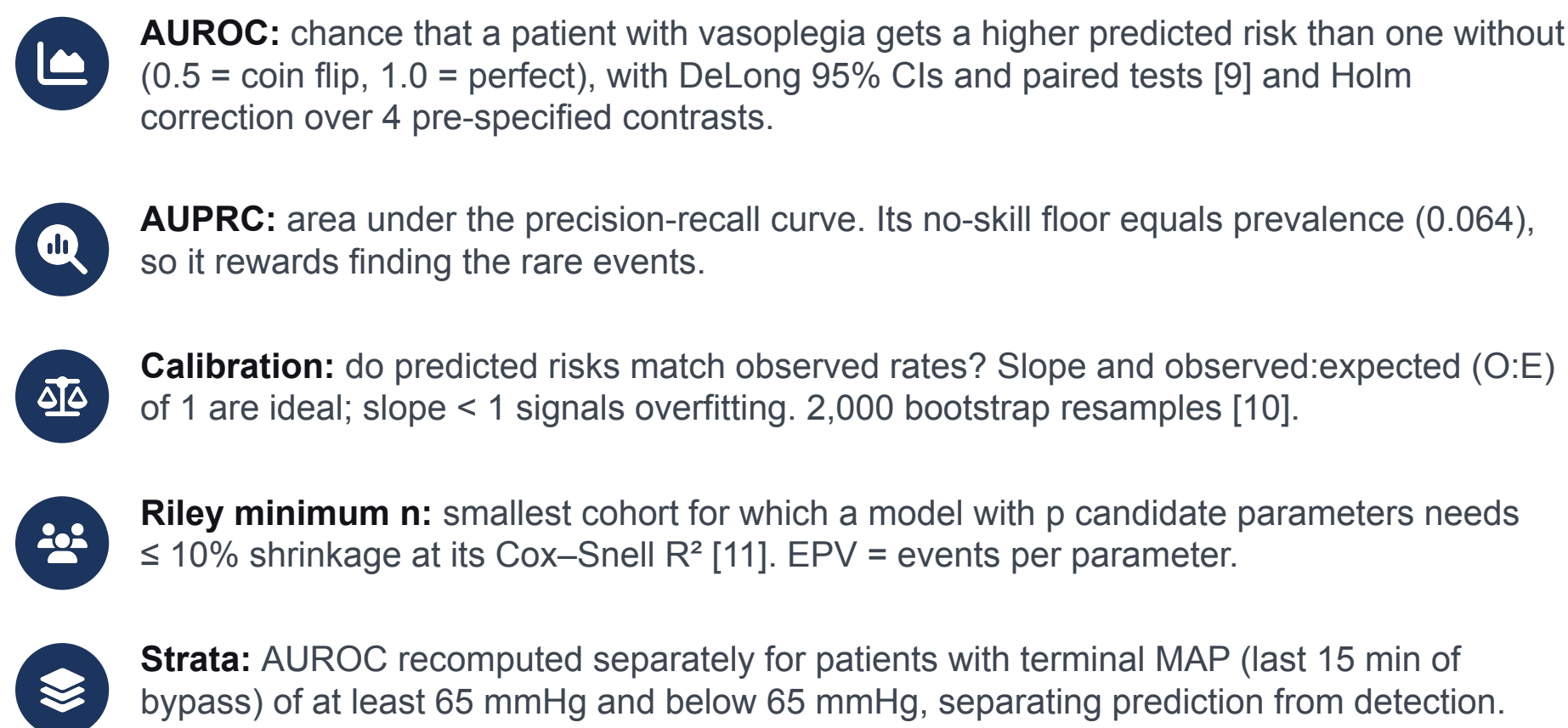
Three-variable model: the predictors the elastic net kept in all 10 folds with the largest standardized coefficients (one from the correlated pressure block), refit by unweighted maximum likelihood.

Nested cross-validation, scored strictly out of fold



Rotate the test fold 10 times → **pooled out-of-fold predictions for all 704 patients** → AUROC, AUPRC, calibration. The three-variable model used the same outer folds. Key comparisons repeated over **20 random re-partitions** with a fixed-hyperparameter elastic net (single-partition AUROC 0.848). Reporting follows the TRIPOD+AI statement for prediction models [8].

5 Statistical Analysis

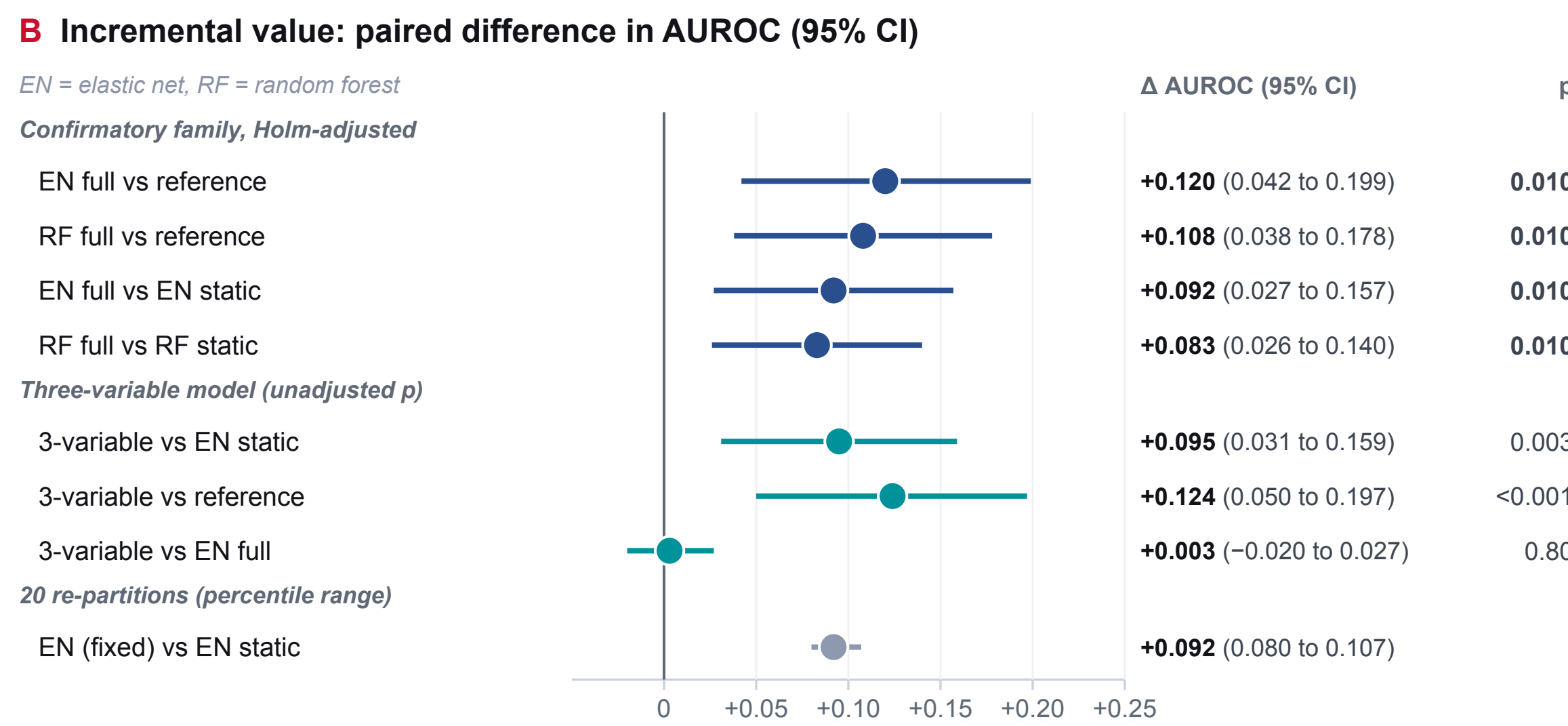
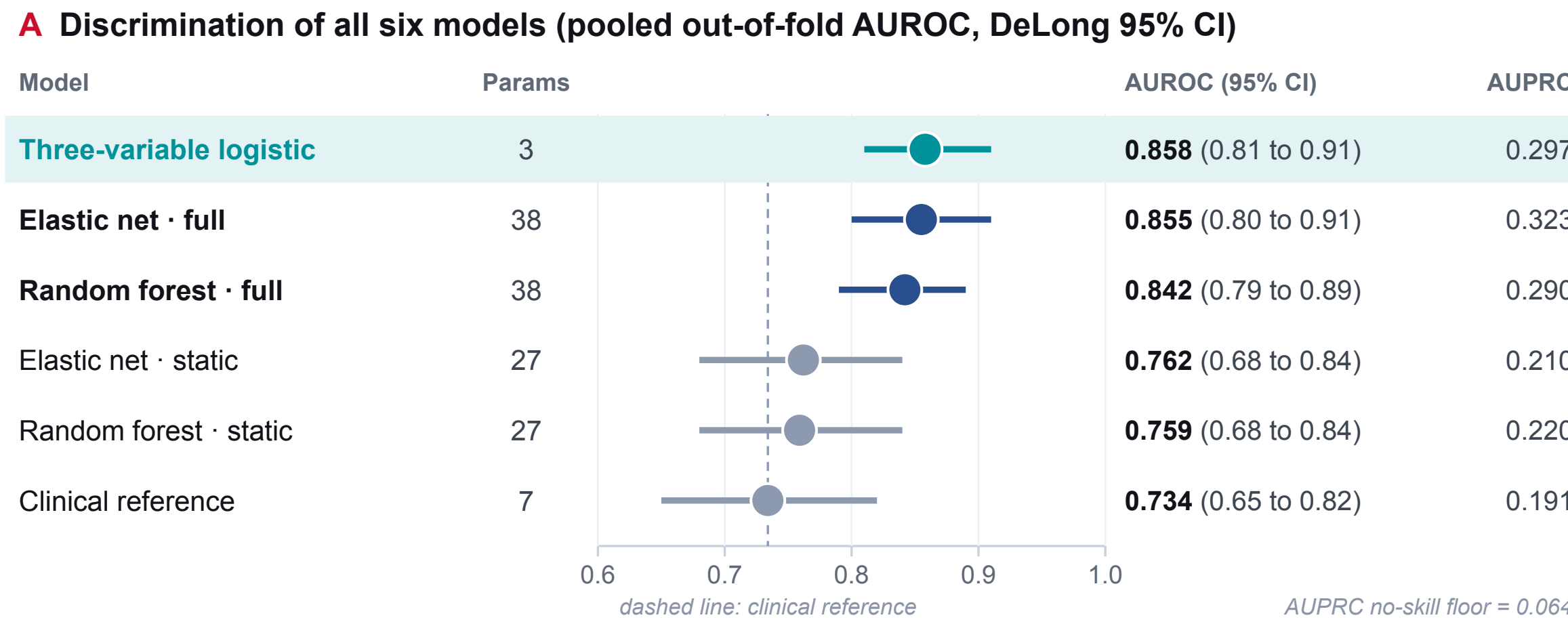


Pre-specified robustness checks (pooled out-of-fold AUROC, DeLong 95% CI)

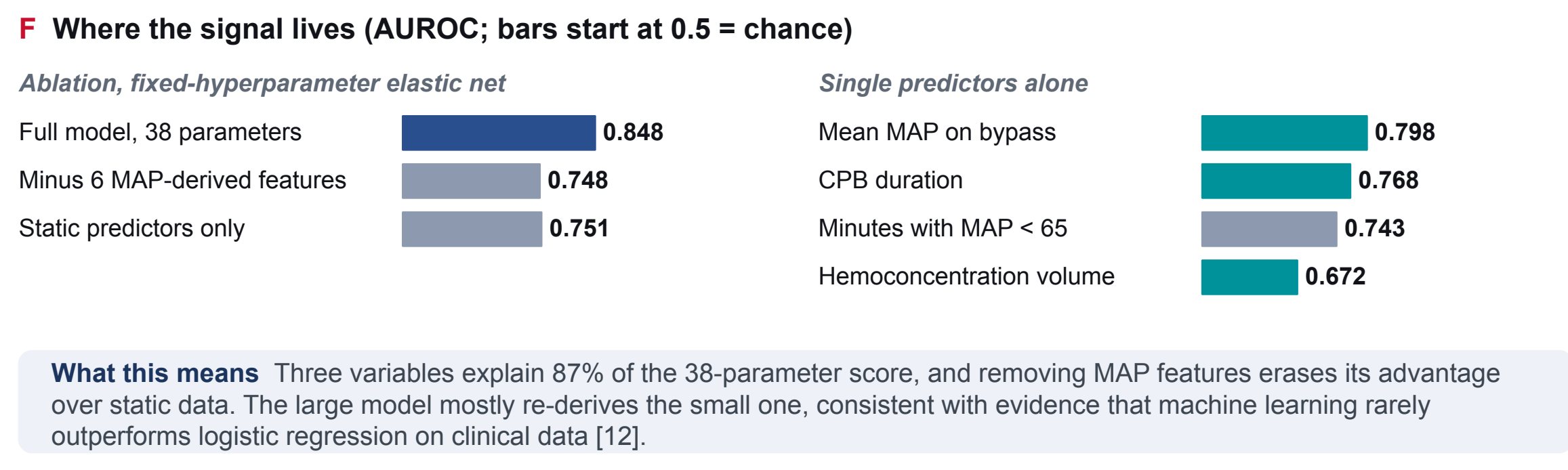
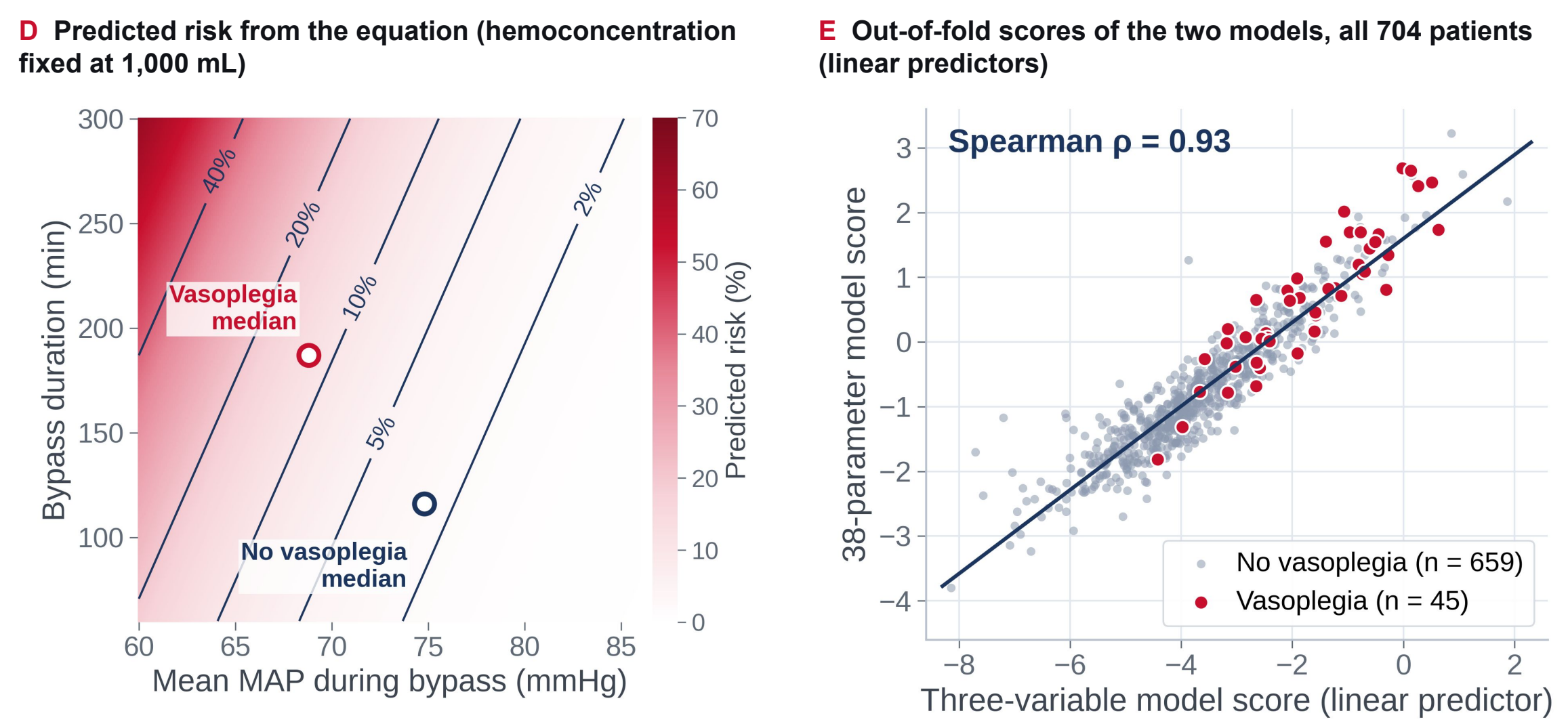
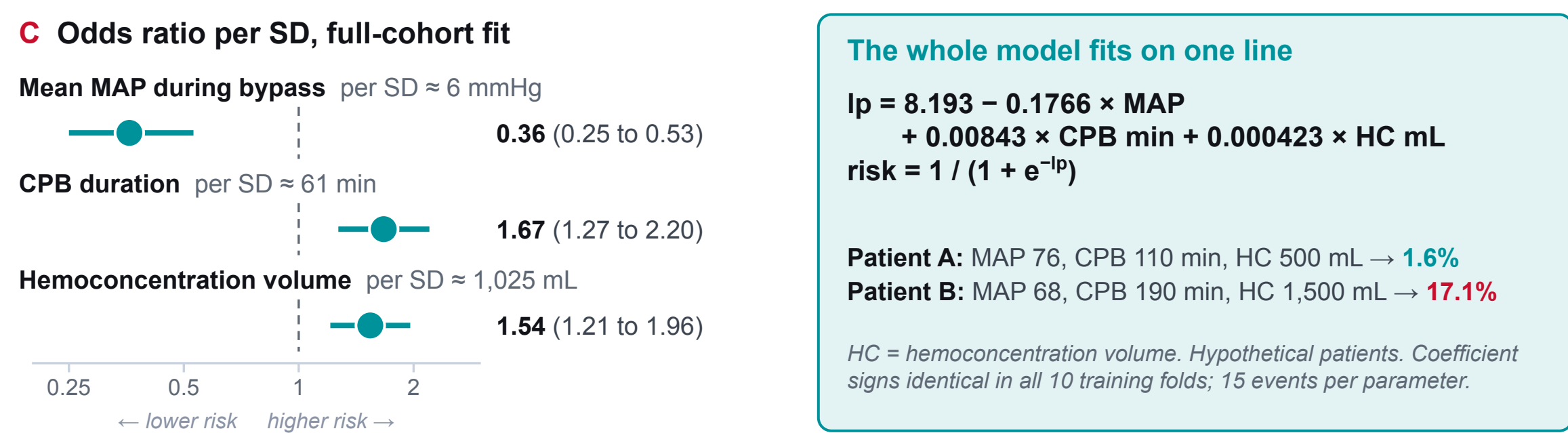
Terminal MAP in place of mean bypass MAP	0.849 (0.79 to 0.91)
Both MAP summaries (four variables)	0.874 (0.82 to 0.92)
Six stable predictors	0.869 (0.82 to 0.92)
Three-variable model with class weighting	0.855
Outcome at MAP < 60 mmHg (16 events), three-variable	0.876 (0.79 to 0.96); fold mean 0.89
Outcome at MAP < 60 mmHg, full elastic net	pooled 0.811; fold mean 0.88
Full model without catheter indicator	0.848
Full model with ambiguous-timing totals	0.864

Full-model rows use the fixed-hyperparameter elastic net, except the MAP < 60 row (nested-tuned), whose lower pooled value reflects probability-scale shifts across folds at 16 events rather than worse ranking.

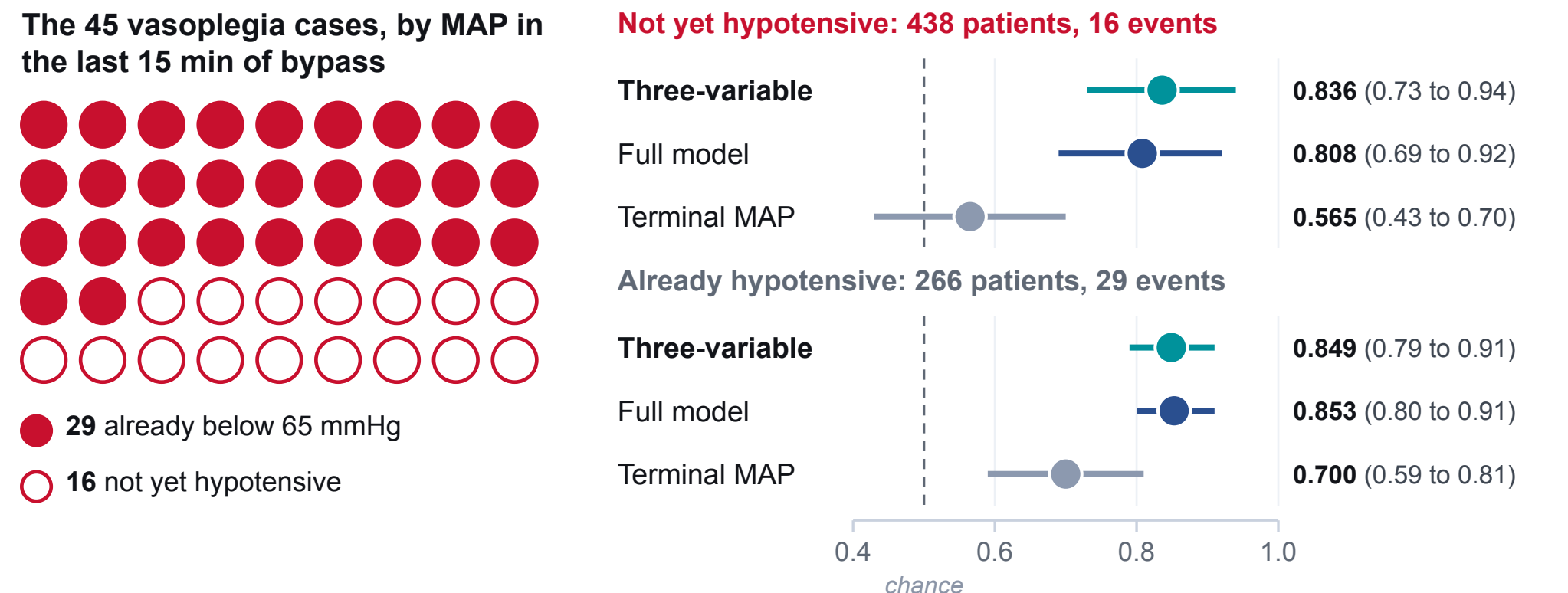
6 Model Performance



7 Three-Variable Model

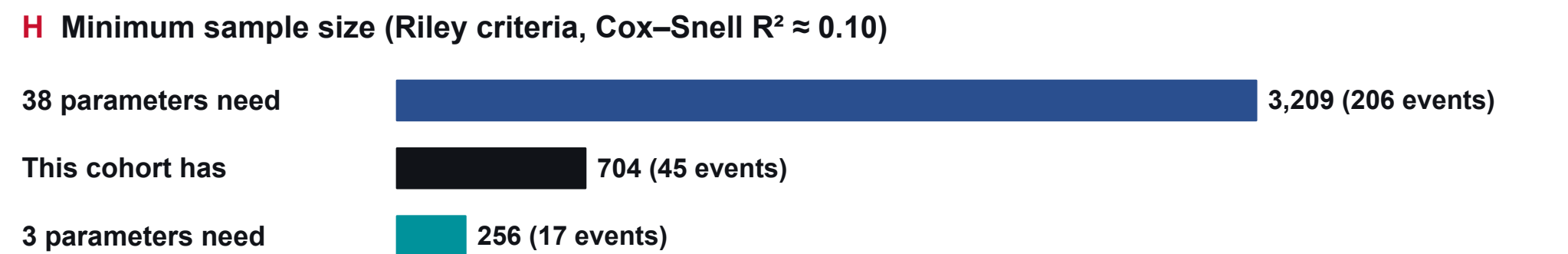
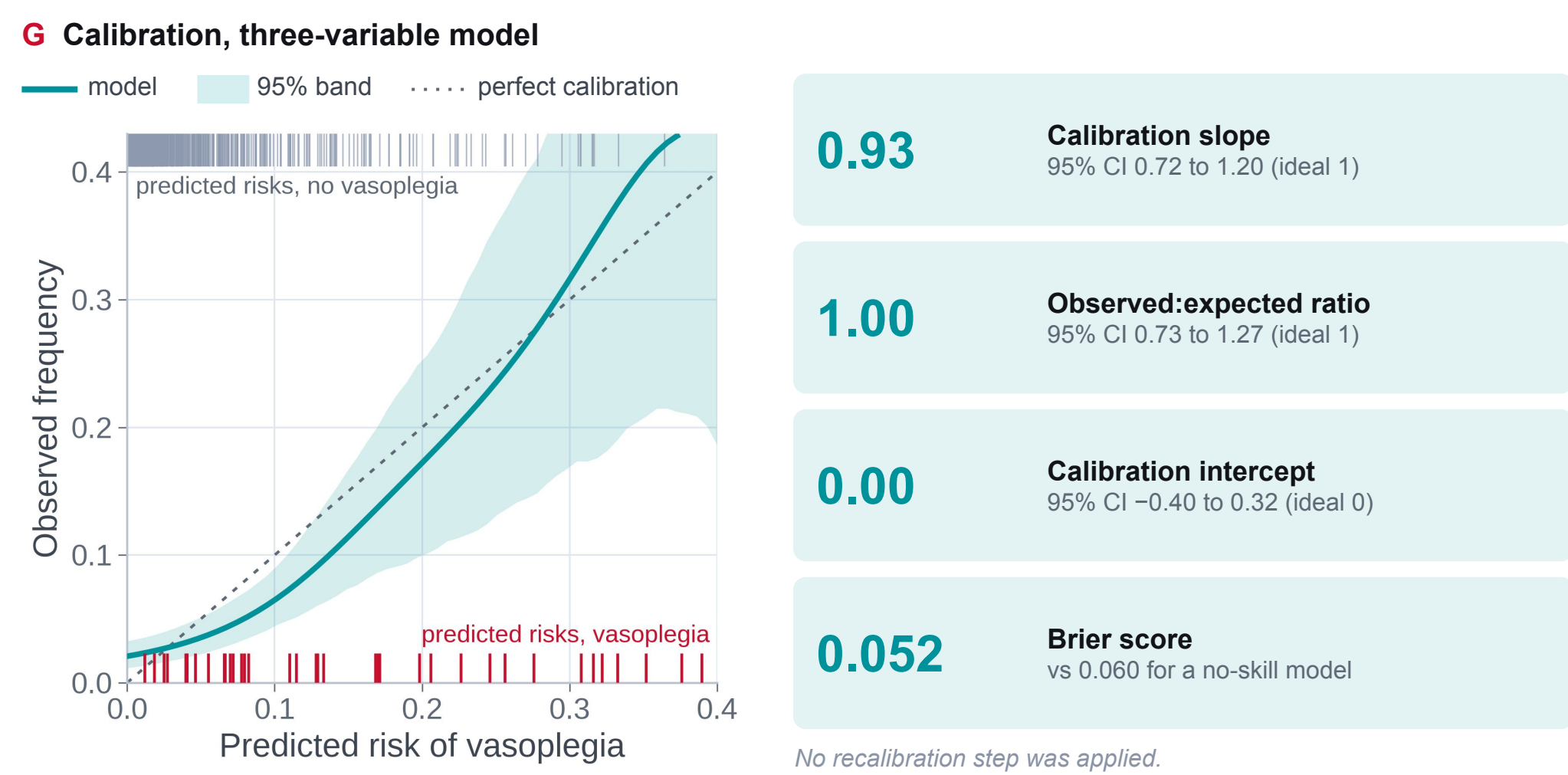


8 Subgroup Analysis



What this means 29 of 45 events were already hypotensive before separation, so a model could score well just by reading the monitor [13]. Where pressure still looked normal, terminal MAP alone was near chance (0.57) while the three-variable model held at 0.84, because it reads accumulated pressure exposure and case burden rather than the current value.

9 Calibration and Sample Size



4.6-fold short for 38 parameters (EPV 1.2) and **2.8-fold above** the need for 3 (EPV 15), if the three are treated as pre-specified, which selecting them from 38 candidates does not fully justify. Cox-Snell R² was 0.101 for the full model (cross-fitted, recalibrated) and 0.100 for the three-variable model.

10 Limitations and Future Work

Selection: the three predictors were chosen once using all 704 patients before out-of-fold scoring, so 0.858 may be optimistic and the 3-parameter Riley figure assumes pre-specification; nested selection is planned. **Inference:** p = 0.80 means no detectable difference, not formal equivalence; DeLong intervals on pooled predictions omit refitting and partition variance. **Scope:** single-center retrospective study without external validation; 45 events overall, and only 16 among patients not yet hypotensive at separation. **Data:** counts are provisional until a few screened records still under curation are resolved; cardiac index is missing in 48% and preoperative lactate in 69%. **Label:** pressure based without a vasopressor criterion; at a 60 mmHg threshold only 16 events remain.



Expanded cohort: a 3- or 5-year total window would add about 787 (= 50 events) or 1,781 (= 114 events) new patients; the final size will follow a validation-specific calculation [15].

Conclusion Three numbers on every perfusion record, mean bypass MAP, bypass duration, and hemoconcentration volume, predicted vasoplegia at separation from CPB as well as a 38-parameter machine learning model, with calibrated risks and within Riley sample-size criteria for a three-parameter model. The next step is validating the frozen model in new patients.

References Acknowledgments: The Tensor Lab for Computational Medicine; UMSOM Office of Student Research.

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